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$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan\left(\frac{x}{2}\right) - 1}{\cos x}$$

$$f(x) = \tan\left(\frac{x}{2}\right) - 1$$

$$f'(x) = \frac{1}{2} \sec^2\left(\frac{x}{2}\right)$$

$$g(x) = \cos x$$

$$g'(x) = -\sin x$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{2} \sec^2\left(\frac{x}{2}\right)}{-\sin x}$$

$$\sec^2\left(\frac{\pi}{4}\right) = 2, \sin\left(\frac{\pi}{2}\right) = 1$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{2} \cdot 2}{-1} = -1$$

References